

Prithvi Raj

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Education

Indian Institute of Information Technology, Guwahati
B.Tech in Electronics and Communication Engineering

Aug 2023 – Current
CGPA: 7.50

Research & Core Projects

Photorealistic 3D Scene Reconstruction using 3D Gaussian Splatting

Oct 2025 – Nov 2025

- Built a 3DGS pipeline to reconstruct traversable 3D scenes from 2D video, creating physics-consistent environments for robot path-planning simulations.
- Optimized the reconstruction process to achieve high-quality visual fidelity with lower computational overhead compared to traditional NeRFs.

Simulation of 6-DOF Robotic Manipulator in Constrained Environments [Github]

Aug 2025

- Orchestrated a complete autonomy pipeline for a mobile manipulator to perform scene-aware grasping and organization tasks, where target objects were randomly initialized in PyBullet/MuJoCo environments.
- Deployed the ROS Nav2 stack, configuring it to use a world model for intelligent navigation, allowing the robot to reliably pathfind to target search areas and object locations.

Multi-agent Swarm Dynamics using FIREBIRD VI [Github]

Jan 2026 – May 2026

UNDER THE GUIDANCE OF DR. SURAJIT PANJA, ASSOCIATE PROFESSOR, IIIT GUWAHATI

- Engineered a decentralized control framework using **ROS2** and **Python** to orchestrate coordinated movement across multiple agents, eliminating the need for a central server.
- Implemented graph-theoretic consensus algorithms to achieve real-time leader-follower formations and autonomous collision avoidance in dynamic environments.

Experience & Leadership

Semiconductor Workforce Development Program TCFST & NTHU, Taiwan

March 2026

- Successfully executed **2-micrometer precision multi-layer fabrication** on a silicon wafer through hands-on cleanroom operations at National Tsing Hua University (NTHU).
- Applied advanced Photolithography, Etching, and Deposition techniques in an industrial-grade fabrication environment as part of the SPARC Indo-Taiwan collaboration.

Research Intern (SPARC Program) IIT Hyderabad

Nov 2025

- Fabricated a functional **Resistive Gas Sensor** detecting different gas (specifically for Ammonia NH_3) levels.
- Successfully executed **multi-layer** semiconductor fabrication with **10-micrometer** detail.
- Studied the end-to-end VLSI manufacturing process, bridging the gap between hardware design and physical silicon constraints to optimize sensor sensitivity.

Coordinator Alankrit Fine-Arts Club – IIIT Guwahati

Dec 2025 – Present

- Led 5+ campus-wide arts events, managing logistics and volunteer teams; recognized as the club's most active coordinator for fostering artistic expression.

Technical Skills

Languages: Python, C++, C, Verilog

Robotics: ROS/ROS2, PyBullet, MuJoCo, Gazebo, Motion Planning, Kinematics

AI & Vision: Scikit-learn, PyTorch/Keras, OpenCV, NeRF, 3D Gaussian Splatting (3DGS)

Hardware: PCB Design (Altium), Embedded Systems, FPGA (Xilinx Vivado)

Tools & Platforms: Git, GitHub, Google Cloud (Gemini API)

Achievements

Hackathon Finalist, India International Science Festival (IISF)

2024

- Recognized as a finalist team in the top 0.2% (from 45,000+ competing teams) in the national-level S&T Hackathon.

First Prize, GitHub University Blind Coding Challenge

2023

- Ranked **#1** among all first-year students across every engineering discipline in a university-wide competition.